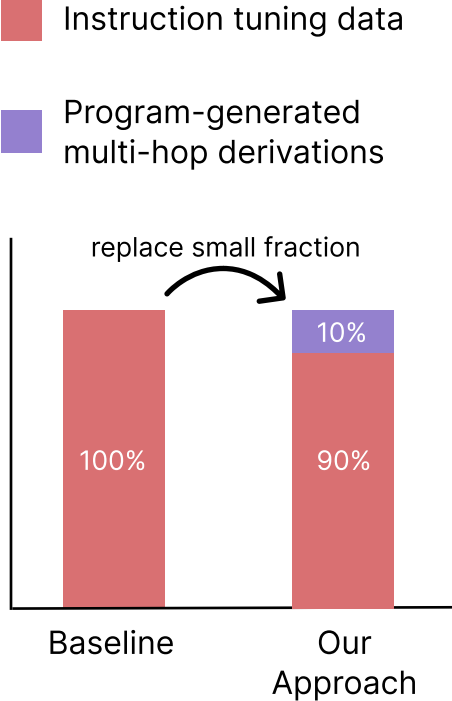


Idea



Example of Program-generated Data Point

Prompt c3 is ruby. c3 is north. c7 is olive.
If c3 is north and c3 is ruby, then c9 is teal.
If c9 is teal, then c9 is west.
If c9 is west and c7 is olive, then c4 is amber.
What is c4?

Latent Proof Rc3, Nc3 $\Rightarrow$ Tc9 $\Rightarrow$ Wc9, with Oc7 $\Rightarrow$ Ac4

Formal Target

<premises>
R c3 ; N c3 ; O c7
N c3 & R c3 $\rightarrow$ T c9
T c9 $\rightarrow$ W c9
W c9 & O c7 $\rightarrow$ A c4
</premises>
<proof> R c3 ; premise
N c3 ; premise
N c3 & R c3 ; &l,2,1
T c9 ; $\rightarrow$ E,3
W c9 ; $\rightarrow$ E,4
O c7 ; premise
W c9 & O c7 ; &l,5,6
A c4 ; $\rightarrow$ E,7 </proof>
<answer> amber </answer>

English Target

<think>
c3 is ruby and c3 is north.
Since c3 is north and c3 is ruby, c9 is teal.
Since c9 is teal, c9 is west.
c7 is olive.
Since c9 is west and c7 is olive, c4 is amber.
Therefore, c4 is amber.
</think>
<answer> amber </answer>

Downstream Performance

